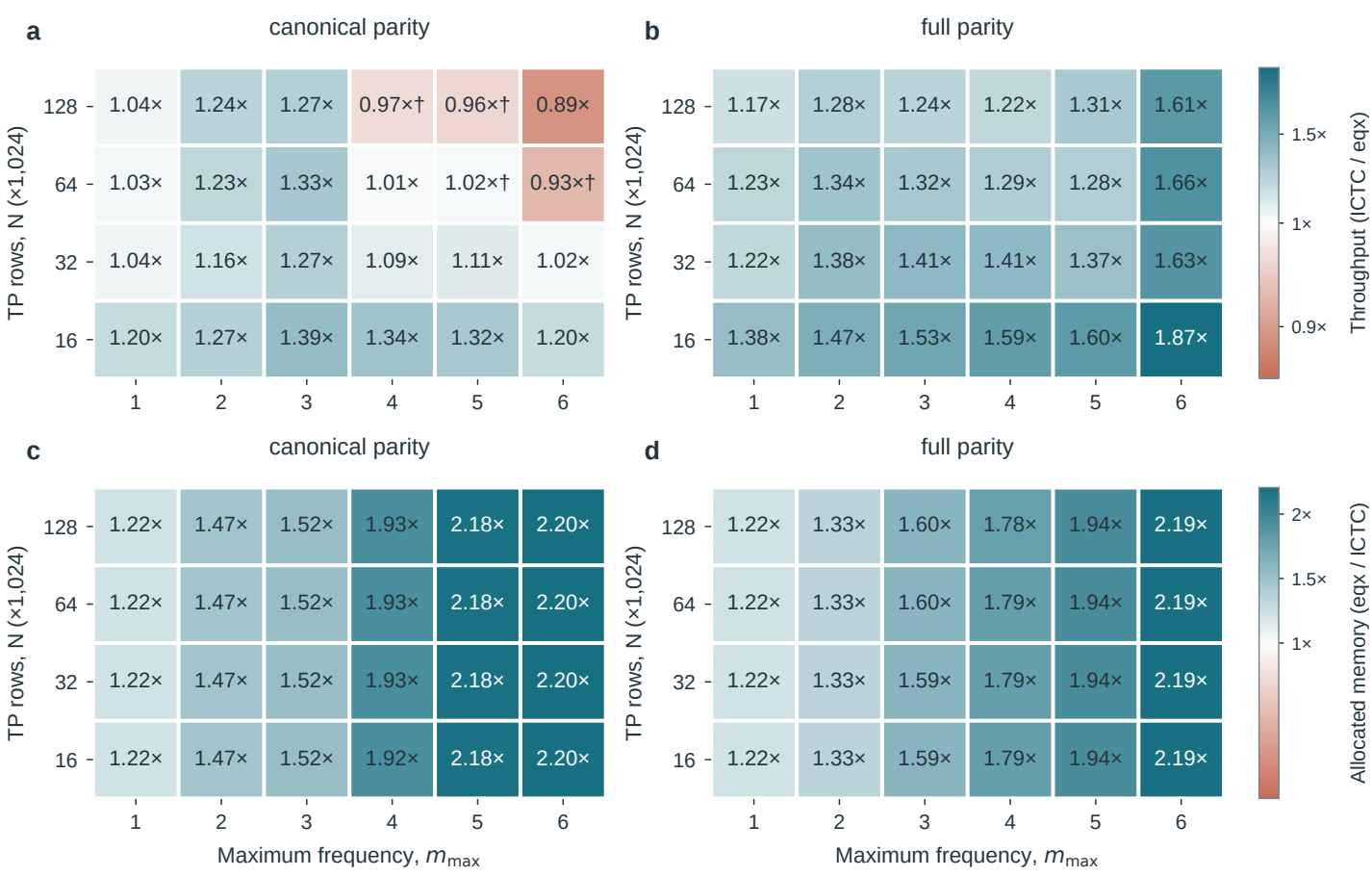

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward + backward · C = 32 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.04× 1.24× 1.27× 0.97×† 0.96×† 0.89×
1.03× 1.23× 1.33× 1.01× 1.02×† 0.93×†
1.04× 1.16× 1.27× 1.09× 1.11× 1.02×
1.20× 1.27× 1.39× 1.34× 1.32× 1.20×
1.17× 1.28× 1.24× 1.22× 1.31× 1.61×
1.23× 1.34× 1.32× 1.29× 1.28× 1.66×
1.22× 1.38× 1.41× 1.41× 1.37× 1.63×
1.38× 1.47× 1.53× 1.59× 1.60× 1.87×
1.22× 1.47× 1.52× 1.93× 2.18× 2.20×
1.22× 1.47× 1.52× 1.93× 2.18× 2.20×
1.22× 1.47× 1.52× 1.93× 2.18× 2.20×
1.22× 1.47× 1.52× 1.92× 2.18× 2.20×
1.22× 1.33× 1.60× 1.78× 1.94× 2.19×
1.22× 1.33× 1.60× 1.79× 1.94× 2.19×
1.22× 1.33× 1.59× 1.79× 1.94× 2.19×
1.22× 1.33× 1.59× 1.79× 1.94× 2.19×
1 2 3 4 5 6
Maximum frequency, m_max
Throughput (ICTC / eqx)
1.5×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5